Chapter 7. Function Spaces

The space C(X), Arzelà–Ascoli, Weierstrass approximation, sequence spaces, improper integrals, spaces of integrable functions

Contents

7.1 The space of continuous functions **2**

7.2 Sequence spaces **3**

7.3 Improper integrals **3**

7.4 Spaces of integrable functions **4**

It is fruitful to regard functions as points of a vector space with a norm. Convergence of functions then becomes convergence of points, and questions such as "does every Cauchy sequence converge?" and "which sets are compact?" make sense.

7.1 The space of continuous functions

A **norm** on a real or complex vector space V is a function $\|\cdot\|$ with $\|v\| \geq 0$ (equality iff $v = 0$), $\|cv\| = |c|\|v\|$ and $\|v + w\| \leq \|v\| + \|w\|$. A normed space is **complete** (a **Banach space**) if every Cauchy sequence converges. For compact $X \subset \mathbb{R}^n$, $C(X)$ denotes the continuous functions $X \to \mathbb{R}$ (or $\mathbb{C}$) with $\|f\|_\infty = \max_X |f|$; convergence in this norm is uniform convergence.

Theorem 7.1

$C(X)$ is complete.

Proof. A Cauchy sequence converges uniformly by Theorem 6.2, and its limit is continuous by Theorem 6.1. ∎

Closed bounded subsets of $C(X)$ need not be compact: $f_n(x) = x^n$ in $C[0,1]$ is bounded but has no uniformly convergent subsequence. The missing ingredient is equicontinuity.

Definition (Equicontinuity)

A family $\mathcal{F} \subset C(X)$ is **equicontinuous** if for every $\epsilon > 0$ there is $\delta > 0$ such that $|f(x) - f(y)| < \epsilon$ whenever $\|x - y\| < \delta$ and $f \in \mathcal{F}$.

A uniformly convergent sequence in $C(X)$ is equicontinuous (X compact): use uniform continuity of finitely many f_n and the uniform closeness of the rest to the limit. A family with uniformly bounded derivatives (or a common Lipschitz constant) is equicontinuous.

Theorem 7.2 (Arzelà–Ascoli)

If X is compact and $\mathcal{F} \subset C(X)$ is bounded and equicontinuous, then every sequence in $\mathcal{F}$ has a uniformly convergent subsequence.

Proof. Let $E = \{e_1, e_2, \dots\}$ be a countable dense subset of X (points with rational coordinates near X, suitably chosen). Since $(f_n(e_1))$ is bounded, a subsequence converges at e_1; from it extract a subsequence converging at e_2; and so on. The diagonal sequence g_k (the k-th function of the k-th subsequence) converges at every e_j. Given ϵ, take δ from equicontinuity; by compactness finitely many $e_{j_1}, \dots, e_{j_p}$ have every $x \in X$ within δ of one of them. Then $|g_k(x) - g_l(x)| \leq |g_k(x) - g_k(e_j)| + |g_k(e_j) - g_l(e_j)| + |g_l(e_j) - g_l(x)| < 3\epsilon$ for k, l large, uniformly in x. ∎

Hence a closed, bounded, equicontinuous subset of $C(X)$ is compact. A typical application: solutions of $y' = F(t, y)$ with continuous F exist (Peano's theorem), because approximate solutions form an equicontinuous family.

Theorem 7.3 (Weierstrass approximation)

Every continuous $f : [a, b] \to \mathbb{R}$ is a uniform limit of polynomials.

Proof. Rescale to $[0, 1]$. The Bernstein polynomials $B_n f(x) = \sum_{k=0}^{n} f(\frac{k}{n}) \binom{n}{k} x^k (1-x)^{n-k}$ work: with $p_k(x) = \binom{n}{k} x^k (1-x)^{n-k}$, $\sum p_k = 1$, $\sum \frac{k}{n} p_k = x$, $\sum (\frac{k}{n} - x)^2 p_k = \frac{x(1-x)}{n} \leq \frac{1}{4n}$ (binomial mean

and variance). Given ϵ, take δ from uniform continuity and $M = \|f\|_\infty$; then

$$|B_n f(x) - f(x)| \le \sum_{|k/n - x| < \delta} \epsilon p_k + \sum_{|k/n-x|\ge\delta} 2M p_k \le \epsilon + \frac{2M}{4n\delta^2} < 2\epsilon$$

for large n, by Chebyshev's inequality. ∎

So polynomials are dense in $C[a,b]$; since polynomials with rational coefficients are dense too, $C[a,b]$ has a countable dense set (it is *separable*).

7.2 Sequence spaces

For sequences $a = (a(n))$ define

$$\|a\|_1 = \sum |a(n)|, \qquad \|a\|_2 = \Big(\sum |a(n)|^2\Big)^{1/2}, \qquad \|a\|_\infty = \sup |a(n)|,$$

and $\ell^1, \ell^2, \ell^\infty$ as the spaces where these are finite; c_0 is the space of sequences tending to 0 (with $\|\cdot\|_\infty$). Then $\ell^1 \subsetneq \ell^2 \subsetneq c_0 \subsetneq \ell^\infty$. The triangle inequality in ℓ^2 comes from the Cauchy–Schwarz inequality $\sum |a(n)b(n)| \le \|a\|_2 \|b\|_2$, and $\langle a, b\rangle = \sum a(n)\overline{b(n)}$ is an inner product.

Theorem 7.4

ℓ^1, ℓ^2, ℓ^∞ and c_0 are complete.

Proof. For ℓ^2: if (a_m) is Cauchy, each coordinate sequence $(a_m(n))_m$ is Cauchy, with limit $a(n)$. For any N, $\sum_{n\le N} |a_m(n) - a_k(n)|^2 < \epsilon^2$ for $m, k \ge M$; let $k \to \infty$ and then $N \to \infty$ to get $\|a_m - a\|_2 \le \epsilon$, which also shows $a \in \ell^2$. The other cases are identical. ∎

7.3 Improper integrals

If $f \in \mathcal{R}[a,c]$ for every $c < b$ (where $b \le \infty$ or f is unbounded near b), the **improper integral** is $\int_a^b f = \lim_{c\to b-} \int_a^c f$ when the limit exists; similarly at the left endpoint or at interior singular points (each separately).

Proposition 7.5 (Comparison)

If $0 \le f \le g$ and $\int_I g$ converges, so does $\int_I f$. More generally, if $|f| \le g$ and $\int_I g < \infty$, then $\int_I f$ converges (absolute convergence implies convergence).

Proof. For $f \ge 0$, $c \mapsto \int_a^c f$ is increasing and bounded. In general use the Cauchy criterion: $|\int_c^{c'} f| \le \int_c^{c'} g$. ∎

Proposition 7.6 (Integral test)

If $f : [1,\infty) \to (0,\infty)$ is decreasing, then $\sum f(n)$ and $\int_1^\infty f$ converge or diverge together; indeed $\sum_{n=1}^N f(n) - \int_1^N f$ decreases to a finite limit.

Proof. $f(n+1) \le \int_n^{n+1} f \le f(n)$. ∎

This gives $\sum n^{-p} < \infty$ iff $p > 1$, and Euler's constant $\gamma = \lim(\sum_{n\le N} \frac{1}{n} - \log N)$.

Example 7.1

(a) $\int_0^\infty \frac{\sin x}{x}dx$ converges (integrate by parts on $[1,c]$: $\int_1^c \frac{\sin x}{x} = \cos 1 - \frac{\cos c}{c} - \int_1^c \frac{\cos x}{x^2}$), but not absolutely: $\int_{k\pi}^{(k+1)\pi}|\sin x|/x \geq \frac{2}{(k+1)\pi}$.
(b) $\int_0^1 x^{-p}dx < \infty$ iff $p < 1$; $\int_1^\infty x^{-p}dx < \infty$ iff $p > 1$.

Theorem 7.7

If $\int_I|f| < \infty$, then for every $\epsilon > 0$ there is $\delta > 0$ such that $\int_J|f| < \epsilon$ for every subinterval J of length less than δ (absolute continuity of the integral), and there is a compact interval $K \subset I$ with $\int_{I\setminus K}|f| < \epsilon$.

Proof. The second claim is the definition of convergence at the endpoints. For the first, choose K as in the second claim with bound $\epsilon/2$ and let $B = \sup_K|f|$ (f is bounded on compact subintervals of its domain of integrability); then $\int_J|f| \leq B|J| + \epsilon/2$. ■

7.4 Spaces of integrable functions

On an interval I let $\mathcal{R}^1(I)$ be the functions that are (properly or improperly) Riemann integrable together with $|f|$, and $\mathcal{R}^2(I)$ those with $\int_I|f|^2 < \infty$; set

$$\|f\|_1 = \int_I|f|, \qquad \|f\|_2 = \Big(\int_I|f|^2\Big)^{1/2}.$$

These are seminorms: $\|f\| = 0$ does not force $f = 0$ (change f at one point). Identifying functions with $\|f - g\| = 0$ gives normed spaces, and $\langle f, g\rangle = \int_I f\overline{g}$ is an inner product on $\mathcal{R}^2(I)$.

Proposition 7.8

A normed space is complete iff every absolutely convergent series $\sum v_k$ ($\sum\|v_k\| < \infty$) converges.

Proof. If complete, the partial sums are Cauchy. Conversely, given a Cauchy sequence choose $n_1 < n_2 < \dots$ with $\|v_{n_{k+1}} - v_{n_k}\| < 2^{-k}$; the telescoping series converges, so a subsequence converges, and then the whole Cauchy sequence converges. ■

Example 7.2

$\mathcal{R}^1[0,1]$ is not complete. Let $g(x) = |x|^{-1/2}$ for $0 < |x| \leq 1$ (and $g(0) = 0$), so $\int_{-1}^1 g = 4$, let $r_1, r_2, \dots$ enumerate the rationals in $[0,1]$, and $f_n(x) = \sum_{k\leq n} 2^{-k}g(x - r_k)$. Each $f_n \in \mathcal{R}^1[0,1]$ (finitely many integrable singularities) and $\|f_n - f_m\|_1 \leq 4\sum_{k>m} 2^{-k} \to 0$. If some $f \in \mathcal{R}^1[0,1]$ were the limit, f would have to be unbounded near every rational, i.e. on every subinterval, so f could not be Riemann integrable, properly or improperly, on any subinterval. This defect is what the Lebesgue integral repairs (Chapter 10).

Nevertheless nice functions are dense:

Theorem 7.9

For $f \in \mathcal{R}^2(I)$ (resp. $\mathcal{R}^1(I)$) and $\epsilon > 0$ there is a continuous g vanishing outside a compact subinterval with $\|f - g\|_2 < \epsilon$ (resp. $\|f - g\|_1 < \epsilon$). Step functions may be used instead of continuous ones.

Proof. First truncate to a compact subinterval K (Theorem 7.7) and to a bounded function, losing at most ϵ. For bounded $f \in \mathcal{R}(K)$ choose a partition with $U(f,P) - L(f,P) < \epsilon$; the step function s equal to m_i on each piece satisfies $0 \le f - s$ and $\int(f-s) < \epsilon$, hence $\|f-s\|_2^2 \le 2\sup|f| \cdot \epsilon$. Finally replace each jump of s by a steep linear piece, changing the integral by an arbitrarily small amount. ■

Corollary 7.10 (Continuity of translation)

For $f \in \mathcal{R}^p[a,b]$ $(p = 1, 2)$, extended by 0, $\lim_{h\to 0}\|f(\cdot + h) - f\|_p = 0$.

Proof. True for continuous g with compact support by uniform continuity; approximate f by such g and use $\|f(\cdot+h) - f\| \le 2\|f - g\| + \|g(\cdot + h) - g\|$. ■